

What the Satellites Miss: Remote Sensing, Testimony, and the Prosecution of the Siege of Mariupol

By Oleh Tsumbek

Summary

Mariupol is one of the most thoroughly documented atrocities in the history of international criminal law, and most of that documentation was produced from orbit. This article argues that satellite-based evidence, even at its most rigorous, cannot carry the prosecutions Mariupol will eventually require, and that the institution producing the best remote-sensing work on the siege, Yale's Humanitarian Research Lab, has said as much in its own published reports. Three passages in those reports, on the undercounting built into satellite revisit cycles, on the withholding of facility-level data, and on the limits of remote sensing in the filtration system, each point at the same boundary. A fourth signal comes from the opposite direction: the Human Rights Watch and Truth Hounds reconstruction of the siege shows what becomes provable once testimony is gathered systematically. I connect these signals to the legal elements they bear on, chiefly the "failed to prevent or repress" limb of command responsibility after *Bemba* and the disclosure rules that stopped *Lubanga* in 2008, and I draw on my own experience of the siege to describe what only people who were in the city can supply. For several classes of legal claim, testimony is the only route to proof, and it should be collected with the same formality as satellite imagery.

I. Introduction

For nearly two months in early 2022, during the siege of Mariupol, I lived three floors below ground in the western part of Mariupol, in a shelter that held roughly 340 people. I was a high school student. My days had a routine, which is a strange thing to say about a siege: I spent part of most of them helping to operate the shelter's manual ventilation system, and the rest waiting for news, which arrived on foot. There was no phone signal. What we knew about the city above us came from people who had walked over from other buildings and from the few car radios still running.

Almost nothing I know about those weeks is the kind of thing a satellite can see. That is the subject of this article.

Mariupol may be the most thoroughly documented siege in the history of international criminal law, and most of that documentation came from orbit. Within weeks of the Russian encirclement, which was complete by March 2, 2022, Yale's Humanitarian Research Lab was

publishing damage assessments built on commercial very-high-resolution satellite imagery.¹ The strike on Maternity Hospital No. 3 on March 9 and the destruction of the Drama Theater on March 16, with the word ДЕТИ painted on the forecourt in letters large enough to read from space, were confirmed from above before most of the survivors had a chance to tell anyone what they had seen.² By the time the last defenders at Azovstal surrendered on May 20, the satellite record of the city was already the backbone of the case that international institutions had begun to assemble.³ It still is. The Yale study of medical facilities, which found that fifty-six of the city's seventy-three medical facilities, seventy-seven percent, were damaged during the siege, is the kind of evidence prosecutors will reach for first.⁴

I think that record is indispensable, and I also think it cannot carry the case on its own. None of that is a complaint about the people who built it. My claim is that the limits of satellite evidence are written into the satellite evidence itself: the institution that produces the most rigorous remote-sensing documentation of Mariupol has, in three separate reports, told its readers where its method stops. A fourth signal comes from the other direction, from the Human Rights Watch and Truth Hounds reconstruction of the siege, which shows what becomes provable once testimony is collected at scale. And the doctrine makes the gap matter. After the ICC Appeals Chamber's acquittal in *Bemba*, command-responsibility prosecutions turn on whether the accused had a discrete chance to act between one attack and the next, and that is exactly what a satellite chronology tends to blur.

Part II explains why these gaps matter as a matter of law. Part III reads the four signals. Part IV takes the strongest objection seriously. Part V describes, partly from experience, what only local knowledge can supply, and Part VI sets out what a testimonial layer built to evidentiary standards would look like.

One word about method. My family is from Mariupol and I was in the city during the opening phase of the siege. That does not make my reading of the law any better, and I have not treated memory as a substitute for documents. It does give me a vantage point on one specific question: the distance between what the satellites recorded and what the city was. Where my own experience shapes a claim below, I say so.⁵

II. Why the Gaps Matter in Law

Three bodies of doctrine turn the methodological problem into a legal one. Command responsibility first. Under Article 28(a) of the Rome Statute, a military commander is criminally responsible for crimes committed by forces under his effective control if he knew, or should have known, that they were committing or about to commit those crimes, and failed to take all necessary and reasonable measures within his power to prevent or repress them or to refer the matter to the competent authorities.⁶ The second limb is about time. To show that a commander

failed to prevent or repress, the prosecution has to show that there was a moment at which he could have done so: an interval between one wave of attacks and the next in which an order could have been given and was not.

Bemba raised the price of proving that. In 2016 the Trial Chamber convicted Jean-Pierre Bemba under Article 28(a) as a remote commander whose forces committed crimes in the Central African Republic.⁷ In June 2018 a divided Appeals Chamber reversed by three votes to two. Among its reasons was that the Trial Chamber had not assessed with enough specificity what measures Bemba could realistically have taken from a distance, given what he actually knew, and, on the majority's account, had moved too quickly from the pattern of the crimes to the failures of the man.⁸ The reaction from scholars was hostile; Leila Sadat wrote that the four separate opinions had "created considerable confusion over the state of ICC law and procedure."⁹

I read it less dramatically. The majority did not abolish remote-commander liability. It demanded evidence granular enough to distinguish one opportunity to act from the next, which is a high bar and, for Mariupol, a bar that satellite chronology alone cannot clear. I will come back to why.

Second, crimes against humanity. Article 7 requires a widespread or systematic attack directed against a civilian population, pursuant to a state or organizational policy.¹⁰ For the filtration system that Russian and Russian-aligned forces ran in occupied Donetsk oblast, the systematic element depends on how the sites actually worked: who was registered, who was held, who was separated out and on what criteria. The Yale filtration report, discussed below, mapped twenty-one sites from orbit and then said plainly that the count could not be completed from orbit.¹¹

Third, disclosure. Article 67(2) obliges the Prosecutor to disclose to the defense evidence that tends to show the innocence of the accused, to mitigate guilt, or to affect the credibility of prosecution evidence, and Rule 77 gives the defense a right to inspect material in the Prosecutor's possession that is material to the preparation of the defense.¹² Article 54(3)(e) lets the Prosecutor accept documents on condition of confidentiality, solely for the purpose of generating new evidence.¹³ In 2008, in *Lubanga*, Trial Chamber I stayed the proceedings in what would become the Court's first trial, because the Prosecutor had gathered a large body of material under such agreements and could not disclose it to the defense.¹⁴ I raise *Lubanga* because a great deal of the satellite record on Mariupol is, by design, unavailable to anyone outside a small circle of accountability mechanisms. That is a sensible security choice. It is also the same problem that stopped *Lubanga* for months, in a different setting.

III. Four Signals from the Record Itself

Defending the claim that satellite evidence cannot stand alone is harder than it sounds, because the satellite chronologies behind any Mariupol prosecution are not public at the level of individual buildings. What is public is the methodology that Yale's Humanitarian Research Lab chose to publish, in three reports issued between May and August 2022. The reading that follows is deliberately narrow. It does not compare satellite data to survivor testimony; that comparison cannot be made from public sources. It shows, in Yale's own words, that Yale has already drawn the boundary.

A. The Undercount

The first signal is in the Lab's damage assessment for Sievierodonetsk raion, issued June 29, 2022, which catalogues damage to 51 of 89 healthcare and educational facilities between February 24 and June 13.¹⁵ The report is unusually open about its method. Figure 1 of its appendix shows the same facility on three dates: March 29 (no visible damage), May 16 (partial damage), and May 30 (additional partial damage). Forty-eight days separate the first two images. The analysts spell out what that means. "It is not possible to conclude exactly how many separate instances of damage occurred to a facility given two images separated by an extended time period," they write, and the chronological analysis therefore "indicates the minimum number of occasions on which an identified location incurred damage."¹⁶

Read against Article 28(a), that sentence is the whole problem in miniature. A satellite that revisits every few days or weeks records a change of state, and if three strikes fell in the interval it records one. The "failed to prevent or repress" element asks whether the commander had a chance to intervene between strike one and strike three. The image cannot say. Yale is careful to present its counts as minimums, which is the right thing to do scientifically and exactly the wrong form for a post-*Bemba* prosecution, which needs to know how many separate occasions there were. The report itself draws the legal consequence: each attack on a facility, "even the same facility being attacked multiple times across separate incidents," may count as a distinct criminal act.¹⁷

B. The Redaction

The second signal is in the Lab's summary of May 17, 2022, on the bombardment of Ukrainian healthcare facilities. The summary reports its findings only in aggregate: 22 of 277 facilities verified as damaged between February 24 and March 29, 2022, across five cities and regions, with annotated imagery released for just seven of them.¹⁸ It also explains why nothing more granular is public. "Due to the report's granularity and the potential use of sensitive information for targeting purposes," it says, "only this summary and examples of annotated imagery depicting damage to seven (7) of these health facilities . . . are being released publicly. The full report will be made available as needed to international accountability mechanisms."¹⁹

I have no quarrel with the redaction. Publishing the coordinates of damaged hospitals in a city under active bombardment would have been indefensible. But the evidentiary cost has gone largely unremarked. Nobody outside the accountability mechanisms, and that includes defense counsel, independent experts, and scholars, can audit the chain of inferences that runs from a set of satellite observations to the claim of “widespread and systematic bombardment.” At trial, that chain will be attacked, and the defense will be entitled under Article 67(2) and Rule 77 to the material it needs to attack it. If the underlying data was received under Article 54(3)(e) conditions, or under an NGO’s equivalent, the Court will face the *Lubanga* question again: what happens when the foundation of the case cannot be shown to the accused. The answer in 2008 was a stay. The way to avoid repeating it is to design the disclosure protocol now, while the reports are being written, rather than in the courtroom.

C. The Filtration Report

The third signal is the most important, because it concerns the part of the Mariupol record that matters most for crimes against humanity and for any eventual genocide case. The Lab’s August 25, 2022 report on filtration identifies twenty-one sites in Donetsk oblast and maps the roles they play: registration, holding, secondary interrogation, and longer-term detention, with a single site often serving more than one.²⁰ It is candid about scale. Ukraine’s former Ombudswoman for Human Rights put the number of civilians transferred to Russia at 1,700,000 by June 14, 2022, including 276,000 children; the Russian government, on August 16, 2022, claimed twice that.²¹ Then the report says, in its executive summary: “The total number of facilities engaged in filtration operations in Donetsk oblast cannot be conclusively determined by open source analysis and remote sensing alone. Ground investigations are required to cross corroborate these initial findings.”²²

That is Yale’s sentence, not mine. The institution whose imagery underpins much of the public understanding of filtration has written into its foundational report that the system cannot be fully seen from above. For Article 7 purposes the point is sharper still. Overhead imagery can show that a site existed and roughly how busy it was. It cannot show whether men of military age were separated from their families at the gate, or whether people identifiable as Ukrainian speakers were held longer, and it is those facts that turn an ugly administrative process into a systematic attack on a civilian population.

D. The View from the Ground

The fourth signal comes from the opposite direction. In 2024 Human Rights Watch and Truth Hounds, working with SITU Research, published the most thorough attempt so far to combine satellite imagery, open-source material, and survivor testimony into a single account of the siege. The report draws on 240 interviews, more than 850 verified photographs and videos, dozens of satellite images, and 3D reconstructions of seven damaged buildings.²³ The estimate of the dead

shows what the added layer buys. By analyzing satellite imagery and drone footage of five cemeteries in and around Mariupol, and by counting not only the individual graves visible from above but also the plaques and crosses that mark trench-style mass graves, which can only be seen from the ground, the report estimates that at least 10,284 people were buried in the city between March 2022 and February 2023, more than 8,000 of them likely from war-related causes above the city's normal mortality rate.²⁴ The city's then-mayor estimated 22,000 dead. The UN monitoring mission had recorded fewer than 1348 civilian deaths. The three numbers do not contradict each other. The UN figure is what an outside body could verify without entering the city; the mayor's figure is what local officials believed; the HRW figure sits between them and is the only one of the three that could be defended in court, and it could be defended only because the satellite count was combined with knowledge of how the cemeteries were actually being used. A satellite can count graves. It cannot say who is in them, or what has been moved.

The report's case studies make the same point one incident at a time. Its reconstruction of the strike on the apartment building at 98 Mytropolytska Street pairs a satellite chronology of the collapse with testimony from a neighbor who was in the basement when the bomb hit, from people who cleared the site in April, and from a relative who spent months searching the ruins, the makeshift morgue, and the cemeteries for her family.²⁵ The satellite record establishes when the building was hit and how much of it fell. The testimony establishes who was in the basement, who recovered the bodies, and whose names belong to which remains. Neither layer alone would support a charge against the individual or the chain of command responsible.

The same report is also why I think the post-*Bemba* bar can be cleared. Its chain-of command chapter identifies seventeen Russian and Russia-affiliated units that operated in the city, using unit obituaries, official decorations, state media, and verified social-media imagery, and it names ten individuals up the chain of command for whom a command-responsibility analysis is at least *prima facie* viable.²⁶ What the *Bemba* majority wanted was a record in which a remote commander's knowledge and options were facts rather than inferences. The open-source record on Mariupol already has that kind of granularity at the unit level, and it has it because people on the ground and in the diaspora supplied what the satellites could not.

IV. The Boilerplate Objection

The strongest objection to this reading is that I am mistaking hedges for admissions. On that view, the undercount language in the Sievierodonetsk report is ordinary methodological caution; every empirical study contains a sentence like it. The redaction language protects sources rather than concealing weakness. And "ground investigations are required" is a research plan, not a limit on the findings.

Each of those readings has something to it, and I have read these passages closely enough to know that other readings remain open. But one caveat can be dismissed as boilerplate. Three of them, spread across a summer of reports and all pointing at the same boundary between what remote sensing can establish and what requires corroboration on the ground, look like an institutional position. I am not arguing that Yale's reports are weak. They are exactly as strong as they claim to be, and what they claim falls short of what a conviction requires. Yale, to its credit, said so first.

V. What Only Local Knowledge Can Supply

Yale's reports mark the outer edge of what satellite-only work can do. HRW's shows what testimony adds when it is gathered in volume. What neither can show is what testimony is for, from the inside. I want to describe three things the satellites missed in the western districts, because each of them corresponds to a legal problem.

The first is information. Mobile service in our part of the city was effectively gone by the fourth day of the invasion, and the electricity went within a couple of days after that.²⁷ In the institutional record these are outages, two lines in a timeline. What the record does not contain is the system that grew up in the silence. Stairwells became message boards. A person with a charged phone was a courier, walking from building to building and repeating what they had heard, and the few car radios still running became shared property. In the shelter, news arrived at walking speed, sometimes days old, and often contradicted by the next person down the stairs.

Two legal consequences follow, and both bear on how testimony from Mariupol will be judged. The first is credibility. A survivor who is asked, years later, to put the events of March 2022 in order is reconstructing a sequence that was already assembled from fragments at the time. None of us lived it as a continuous timeline. When a defense lawyer finds an inconsistency in a witness's chronology, the inconsistency is usually a product of the information environment rather than a failure of memory, and a tribunal that does not understand the environment will draw the wrong inference. The second is corroboration. The absence of mobile metadata for the first weeks of the siege is itself an evidentiary fact. A court that expects the usual digital trail, timestamps, location data, message logs, will find almost none for those weeks, and it will have to calibrate what corroboration it can reasonably demand.

The second thing the satellites miss is the shape of a day. The Yale medical-facilities study works at the level of the whole city, and at that level its data is rigorous. But the chronology satellite imagery produces, a pre-strike image and a post-strike image separated by however long the satellite took to come back, tends to fold successive events into one. From below, an attack has a beginning and an end, and then another beginning. From above, it is a change between Tuesday and Friday.

That is the Sievierodonetsk undercount as it was lived. For the “failed to prevent or repress” element, the difference between one morning of bombardment and two attacks a day apart is the difference between one opportunity to intervene and two, and a satellite chronology that merges them is weaker evidence on that element than a lived chronology that keeps them apart. Memory is worse than satellite imagery on where and how much. It is better on how many times. The two records need each other, and the way to get them working together is testimony gathered street by street, which can pull apart what overhead imagery has merged.

The third thing is the social structure of filtration. Yale mapped twenty-one sites and the roles each one played. What imagery cannot show is the line between sites that processed civilians on what was essentially an administrative basis and sites that, at particular moments, worked as instruments of selection. The legal difference is large. A filtration system organized administratively, even one that produced torture, supports war-crimes charges. A system in which particular groups were systematically separated for different treatment, men of military age, public servants, people marked as Ukrainian speakers, is a much stronger foundation for crimes against humanity and possibly for genocide. Drawing that line takes testimony organized around what survivors of particular sites can say about who was held, who was released, and on what apparent criteria. The anthropologist Veena Das has written, in another context, about the distance between an event as documented and an event as lived.²⁸ The Mariupol filtration record has exactly that distance in it, and closing it is the most urgent testimonial task the case presents.

VI. Building the Testimonial Layer

If the argument so far is right, testimony from Mariupol should be collected with the same chain-of-custody discipline and the same written protocols that open-source investigators bring to imagery, and with funding that lasts as long as the cases do. Right now it is not. Open source work has Bellingcat’s methods and the Berkeley Protocol on Digital Open Source Investigations, which sets standards for collecting, preserving, and authenticating digital material for use in international criminal proceedings.²⁹ Remote sensing has Yale’s pipeline. Structured testimonial integration has no equivalent. There is no Berkeley Protocol for interviewing a survivor of filtration in a way that will still hold up in front of a hostile cross-examiner in 2034.

Start with a protocol. SITU Research’s 3D reconstructions of seven Mariupol buildings show what one could be built around: a survivor who can point to the basement room in a model, rather than describe it from memory, gives evidence that is easier to test and harder to shake.³⁰ A written standard for testimonial collection in remote-sensing cases would specify how a witness’s account is anchored to the imagery, how discrepancies between the two are recorded rather than smoothed away, and how the testimony is preserved so that a court can later see what was asked and what was answered.

Disclosure needs planning too. Every satellite-based report withheld from the public for security reasons should be accompanied, from the start, by a plan for how its underlying data will reach the defense when the time comes. Article 54(3)(e) exists for material obtained on confidential terms. It does not exist to keep a prosecution's foundation permanently out of reach, and *Lubanga* is the record of what happens when it is used that way.³¹ Deciding now which parts of the Mariupol satellite record can be disclosed, to whom, and under what protective measures, is cheaper than a stay.

And money. Yale's Humanitarian Research Lab lost its federal funding in 2025, nearly closed that summer, and has survived since on individual donations; in September 2026 its director warned that the lab's Ukraine team would be let go around Christmas unless new funding arrived.³² Testimonial work is more exposed still, because it has no institutional home: it is done by NGOs on grants, by national prosecutors as a sideline, and by survivors' own networks for nothing. The International, Impartial and Independent Mechanism for Syria showed that pooled, multi-year funding for evidence work is possible.³³ Something like it for the testimonial layer on Ukraine would be a small expense next to what has already been spent on the satellite one.

VII. Conclusion

A tribunal that reconstructs Mariupol only from remote-sensing data will produce an accurate but partial city. A tribunal that also uses testimony gathered under proper safeguards will produce a city its survivors can recognize. I have argued that the second should be the goal, and that meeting it is a matter of method rather than sentiment: Yale's own reports mark where the satellite record stops, HRW's shows what testimony adds, and the *Bemba* and *Lubanga* lines of doctrine explain why the difference will decide cases.

When I think about what a verdict on Mariupol would have to contain, I think about the shelter. Three floors down, 340 people, air moved by hand, and news that came down the stairs already days old. None of that is in any satellite image of the city, and all of it is what the city was. International criminal law will be judged on whether the people who lived through those weeks recognize its account of them. That is a higher standard than admissibility. For Mariupol, I do not think a lower one is worth having.

About the Author

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Notes

1. YALE SCH. OF PUB. HEALTH, HUMANITARIAN RSCH. LAB, EVIDENCE OF WIDESPREAD AND SYSTEMATIC BOMBARDMENT OF UKRAINIAN HEALTHCARE FACILITIES: SUMMARY (May 17, 2022) [hereinafter YALE HEALTHCARE SUMMARY].
2. AMNESTY INT’L, “CHILDREN”: THE ATTACK ON THE DONETSK REGIONAL ACADEMIC DRAMA THEATRE IN MARIUPOL, UKRAINE (June 30, 2022) (concluding that the strike was “a clear war crime”); Lori Hinnant, Mstyslav Chernov & Vasilisa Stepanenko, *AP Evidence Points to 600 Dead in Mariupol Theater Airstrike*, ASSOCIATED PRESS (May 4, 2022).
3. See HUM. RTS. WATCH & TRUTH HOUNDS, “OUR CITY WAS GONE”: RUSSIA’S DEVASTATION OF MARIUPOL, UKRAINE (2024) [hereinafter OUR CITY WAS GONE] (chronology of the siege).
4. Danielle N. Poole et al., *The Effect of Conflict on Damage to Medical Facilities in Mariupol, Ukraine: A Quasi Experimental Study*, 5 PLOS GLOB. PUB. HEALTH e0003950 (2025) (identifying seventy-three medical facilities and finding fifty-six damaged between February 24 and May 20, 2022, with no correlation between facility size and damage, a pattern the authors describe as “not random” and possibly the result of intentional targeting).
5. My family lived in the western part of the city, in the district historically known as Zhovtnevyi (renamed Tsentralnyi in 2016 as part of decommunization; Russian occupation authorities have reverted to the pre-2016 name). My position is that proximity to events does not exempt an author from ordinary standards of rigor. It obliges the author to say what proximity contributes and what it cannot settle. Where my recollection corroborates documented facts, such as the loss of municipal services in the first days of the siege, I note it in the text. Everywhere else the analysis rests on public sources.
6. Rome Statute of the International Criminal Court art. 28(a), July 17, 1998, 2187 U.N.T.S. 90 [hereinafter Rome Statute].
7. Prosecutor v. Bemba Gombo, Case No. ICC-01/05-01/08, Judgment Pursuant to Article 74 of the Statute (Mar. 21, 2016) (convicting Bemba under Article 28(a) for crimes committed by Mouvement de libération du Congo forces in the Central African Republic between October 2002 and March 2003).
8. Prosecutor v. Bemba Gombo, Case No. ICC-01/05-01/08 A, Judgment on the Appeal of Mr Jean-Pierre Bemba Gombo Against Trial Chamber III’s “Judgment Pursuant to Article 74 of the Statute” (June 8, 2018). The dissent of Judges Monageng and Hofmański runs to 269 pages against an 80-page majority judgment.
9. Leila N. Sadat, *Prosecutor v. Jean-Pierre Bemba Gombo*, 113 AM. J. INT’L L. 353 (2019); see also Diane Marie Amann, *In Bemba and Beyond, Crimes Adjudged to Commit Themselves*, EJIL: TALK! (June 13, 2018) ; Susana SáCouto & Patricia Viseur Sellers, *The Bemba Appeals Chamber Judgment: Impunity for Sexual and Gender-Based Crimes?*, 27 WM. & MARY BILL RTS. J. 599 (2019).
10. Rome Statute, *supra* note 6, art. 7(1), (2)(a).
11. YALE SCH. OF PUB. HEALTH, HUMANITARIAN RSCH. LAB, SYSTEM OF FILTRATION: MAPPING RUSSIA’S DETENTION OPERATIONS IN DONETSK OBLAST (Aug. 25, 2022) [hereinafter YALE FILTRATION REPORT].
12. Rome Statute, *supra* note 6, art. 67(2); Int’l Crim. Ct., Rules of Procedure and Evidence, r. 77, ICC-ASP/1/3 (2002).
13. Rome Statute, *supra* note 6, art. 54(3)(e).

14. Prosecutor v. Lubanga Dyilo, Case No. ICC-01/04-01/06, Decision on the Consequences of Non-Disclosure of Exculpatory Materials Covered by Article 54(3)(e) Agreements and the Application to Stay the Prosecution of the Accused, Together with Certain Other Issues Raised at the Status Conference on 10 June 2008 (June 13, 2008) [hereinafter Lubanga Stay Decision]. The Appeals Chamber upheld the stay in principle in October 2008, and the trial began only after the providers of the material consented to its disclosure to the Chamber and, in redacted or summarized form, to the defense. *See* Prosecutor v. Lubanga Dyilo, Case No. ICC-01/04-01/06 OA 13, Judgment on the Appeal of the Prosecutor Against the Decision of Trial Chamber I Entitled “Decision on the Consequences of Non-Disclosure of Exculpatory Materials Covered by Article 54(3)(e) Agreements and the Application to Stay the Prosecution of the Accused, Together with Certain Other Issues Raised at the Status Conference on 10 June 2008” (Oct. 21, 2008).
15. YALE SCH. OF PUB. HEALTH, HUMANITARIAN RSCH. LAB, DAMAGE ASSESSMENT OF HEALTH AND EDUCATIONAL FACILITIES IN SIEVIERODONETSK RAION, UKRAINE: EVIDENCE OF WIDESPREAD, INDISCRIMINATE, AND PERSISTENT BOMBARDMENT BY RUSSIA AND RUSSIA-ALIGNED FORCES BETWEEN 24 FEBRUARY – 13 JUNE 2022 (June 29, 2022) [hereinafter YALE SIEVIERODONETSK REPORT]. I cite the Sievierodonetsk report for its methodological transparency, not because the Sievierodonetsk record stands in for the Mariupol record; the same imaging constraints apply across the Lab’s Ukraine work.
16. YALE SIEVIERODONETSK REPORT, *supra* note 15, app. I, fig. 1 and accompanying text.
17. *Id.* (Executive Summary).
18. YALE HEALTHCARE SUMMARY, *supra* note 1 (reporting damage verified at 22 of 277 facilities reviewed across Mariupol, Kharkiv, Kyiv Oblast, Chernihiv, and Izyum, roughly eight percent).
19. *Id.* The summary states that the full report was provided to the OSCE Moscow Mechanism and is available to international accountability mechanisms on request.
20. YALE FILTRATION REPORT, *supra* note 11.
21. *Id.* (attributing the figure of 1,700,000 transferred civilians, including 276,000 children, to the former Ukrainian Ombudswoman for Human Rights as of June 14, 2022, and the figure of 3,400,000, including more than 550,000 children, to the Russian government as of August 16, 2022).
22. *Id.* (Executive Summary).
23. OUR CITY WAS GONE, *supra* note 3 (describing 240 interviews, mostly with displaced Mariupol residents, more than 850 verified photographs and videos, dozens of satellite images, and 3D reconstructions and spatial analysis by SITU Research).
24. *Id.* ch. VIII (analyzing grave sites at five cemeteries: Starokrymske, Novotroitske, Vynohradne, Pavlov Street, and Manhush; the report describes the 10,284 figure as a likely significant underestimate of wartime mortality).
25. *Id.* (case study of the March 11, 2022 strike on Mytropolytska Street 98, a five-story residential building serving as a designated shelter).
26. *Id.* ch. X (Russian Chain of Command). The ten named individuals are Vladimir Putin, Sergei Shoigu, Valery Gerasimov, Sergei Rudskoy, Alexander Dvornikov, Viktor Zolotov, Andrei Mordvichev, Ramzan Kadyrov, Adam Delimkhanov, and Denis Pushilin.
27. Author’s recollection, offered under the caveat in note 5 and not as a substitute for documented sources.
28. VEENA DAS, LIFE AND WORDS: VIOLENCE AND THE DESCENT INTO THE ORDINARY (2007).
29. U.N. OFF. OF THE HIGH COMM’R FOR HUM. RTS. & HUM. RTS. CTR., UNIV. OF CAL., BERKELEY, SCH. OF L., BERKELEY PROTOCOL ON DIGITAL OPEN SOURCE

INVESTIGATIONS (2022); *see also* Bellingcat, *Online Open Source Investigation Toolkit* (updated regularly).

30. *See* OUR CITY WAS GONE, *supra* note 3 (3D reconstructions by SITU Research).

31. Lubanga Stay Decision, *supra* note 14.

32. Ray Furlong, *Exclusive: 'A Catastrophic Blow' as US Shuts Unit Investigating War Crimes in Ukraine*, RADIO FREE EUR./RADIO LIBERTY (Mar. 19, 2025); Yale Sch. of Pub. Health, *Individual Donors Help Humanitarian Research Lab Stay Open Until October* (2025) (reporting that the Lab, which had announced it would cease operations on July 1, 2025, could continue until at least October 1, 2025). Alex Raufoglu, Yale Tracking of Ukrainian Children Taken by Russia at Risk as US Reshapes \$25M Program, Radio Free Eur./Radio Liberty (Sept. 9, 2026) (reporting that the Lab's Ukraine researchers were scheduled to be terminated around Christmas unless new funding was found).

33. G.A. Res. 71/248 (Dec. 21, 2016) (establishing the International, Impartial and Independent Mechanism to Assist in the Investigation and Prosecution of Persons Responsible for the Most Serious Crimes under International Law Committed in the Syrian Arab Republic since March 2011, funded through voluntary contributions).